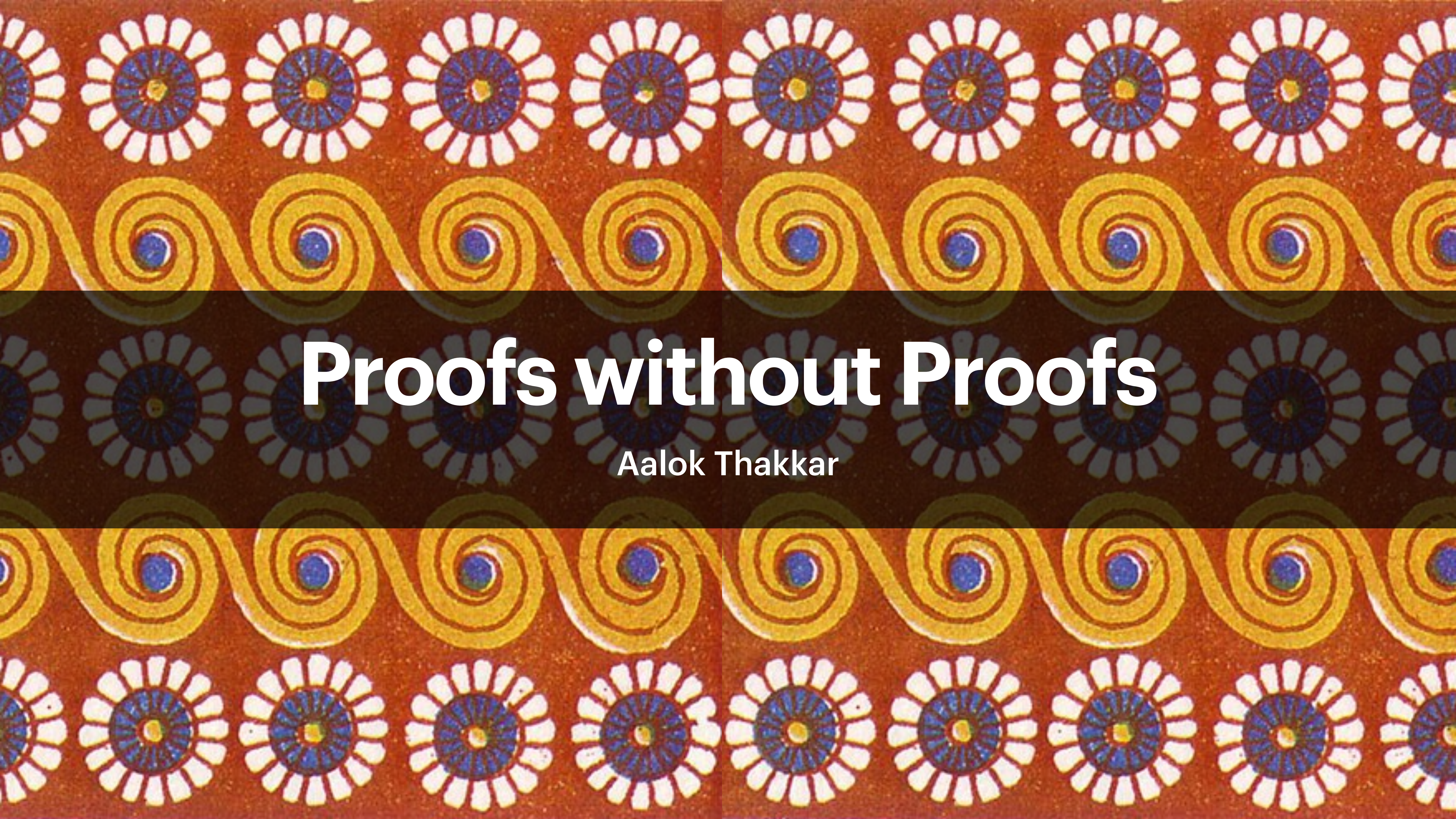




Proofs without Proofs

Aalok Thakkar

The Hindus apparently were attracted by the arithmetical and computational aspects of mathematics rather than by the geometrical and rational features of the subject which had appealed so strongly to the Hellenistic mind. Their name for mathematics, *gaṇita*, meaning literally the ‘science of calculation’ well characterizes this preference. They delighted more in the tricks that could be played with numbers than in the thoughts the mind could produce, so that neither Euclidean geometry nor Aristotelian logic made a strong impression upon them.

M. D. Srinivasa Proofs in Indian Mathematics (2005).

Upapatti

Bhāskarācārya II (c. 1150)

Saṅkara Vāriyar (c. 1535)

Gaṇeśa Daivajña (c. 1545)

Kṛṣṇa Daivajña (c. 1600)

Jyeṣṭhadeva (c. 1530)

And several other works post 1970...

Deficiency View

Formal Greco-
European Proofs of
Correctness

Corrective View

Upapatti in
commentaries

Deficiency View

Formal Greco-
European Proofs of
Correctness

Corrective View

Upapatti in
commentaries

Is there more to *proofs* in Indian Mathematics?

Deficiency View

Formal Greco-
European Proofs of
Correctness

Corrective View

Upapatti in
commentaries

What is the shared assumption between these two perspectives?

Deficiency View

Formal Greco-
European Proofs of
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Corrective View

Upapatti in
commentaries

A proof is a discourse **adjoined to** a
result and **separable from** the operation

Deficiency View

Formal Greco-
European Proofs of
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Corrective View

Upapatti in
commentaries

A proof is a discourse **adjoined to** a
result and **separable from** the operation

In computer science, the proof of correctness is
inseparable from the implementation.

Mathematics is *not changing*.



Mathematics is *not changing*.



Mathematics is *not changing*.

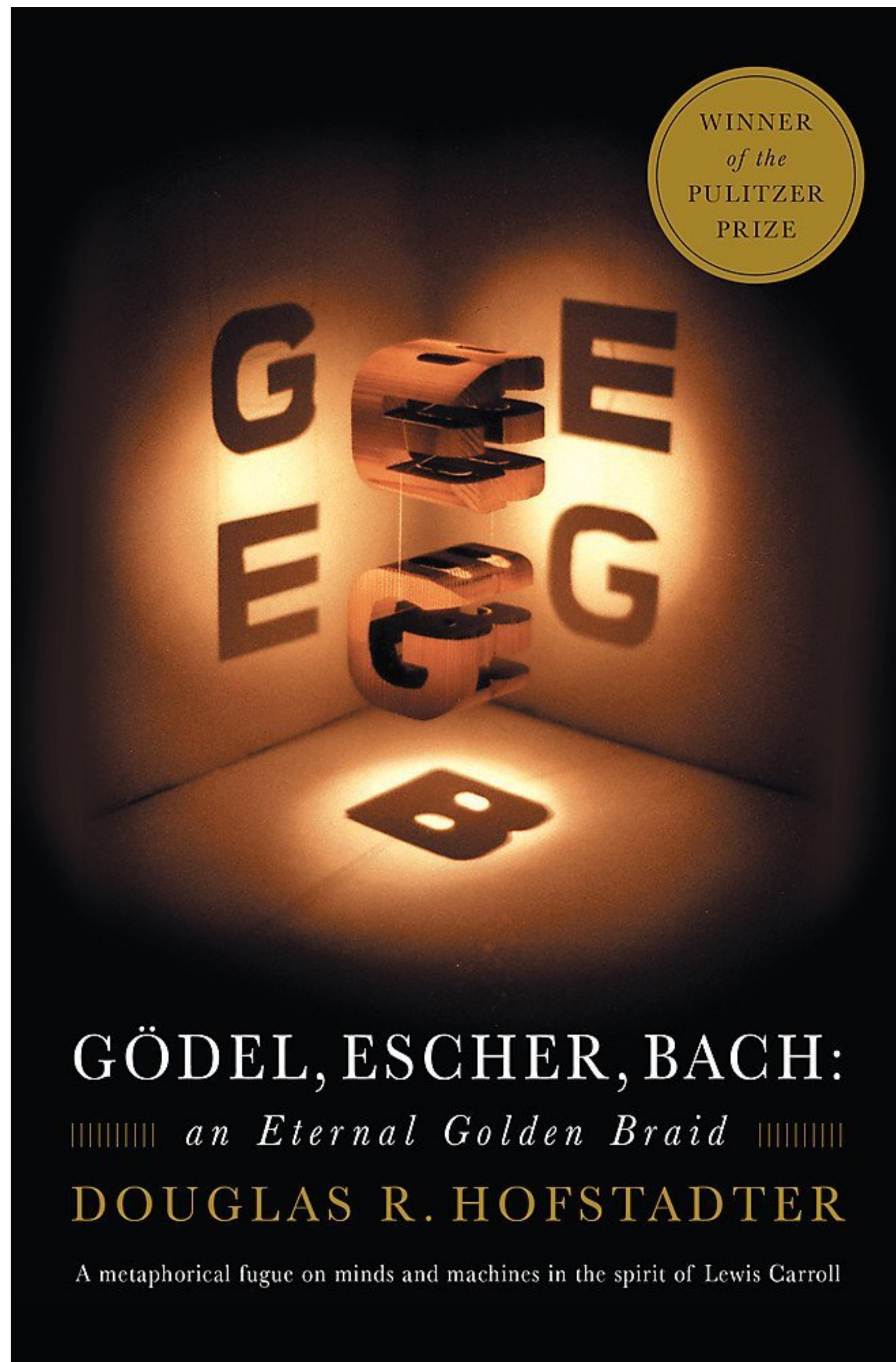


Mathematics is *not changing*.



*Invariant: That which remains
constant under transformations.*



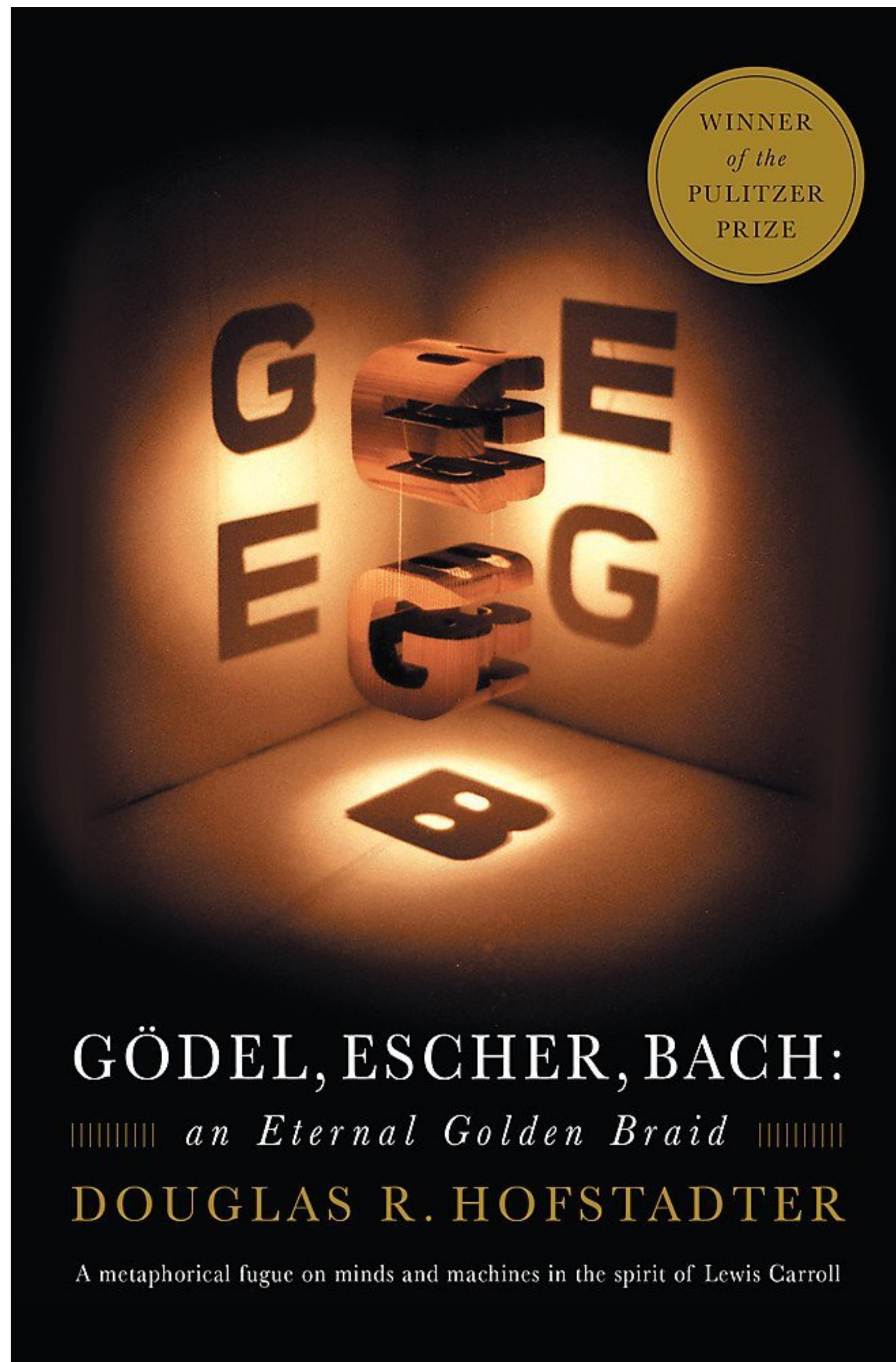


MU Puzzle

We have symbols **M**, **I**, and **U**. Start with string **MI**, and transform using the rules:

1. $xI \rightarrow xIU$
2. $Mx \rightarrow Mxx$
3. $xIIIIy \rightarrow xUy$
4. $xUUy \rightarrow xy$

“Any reachable string must start with **M**”



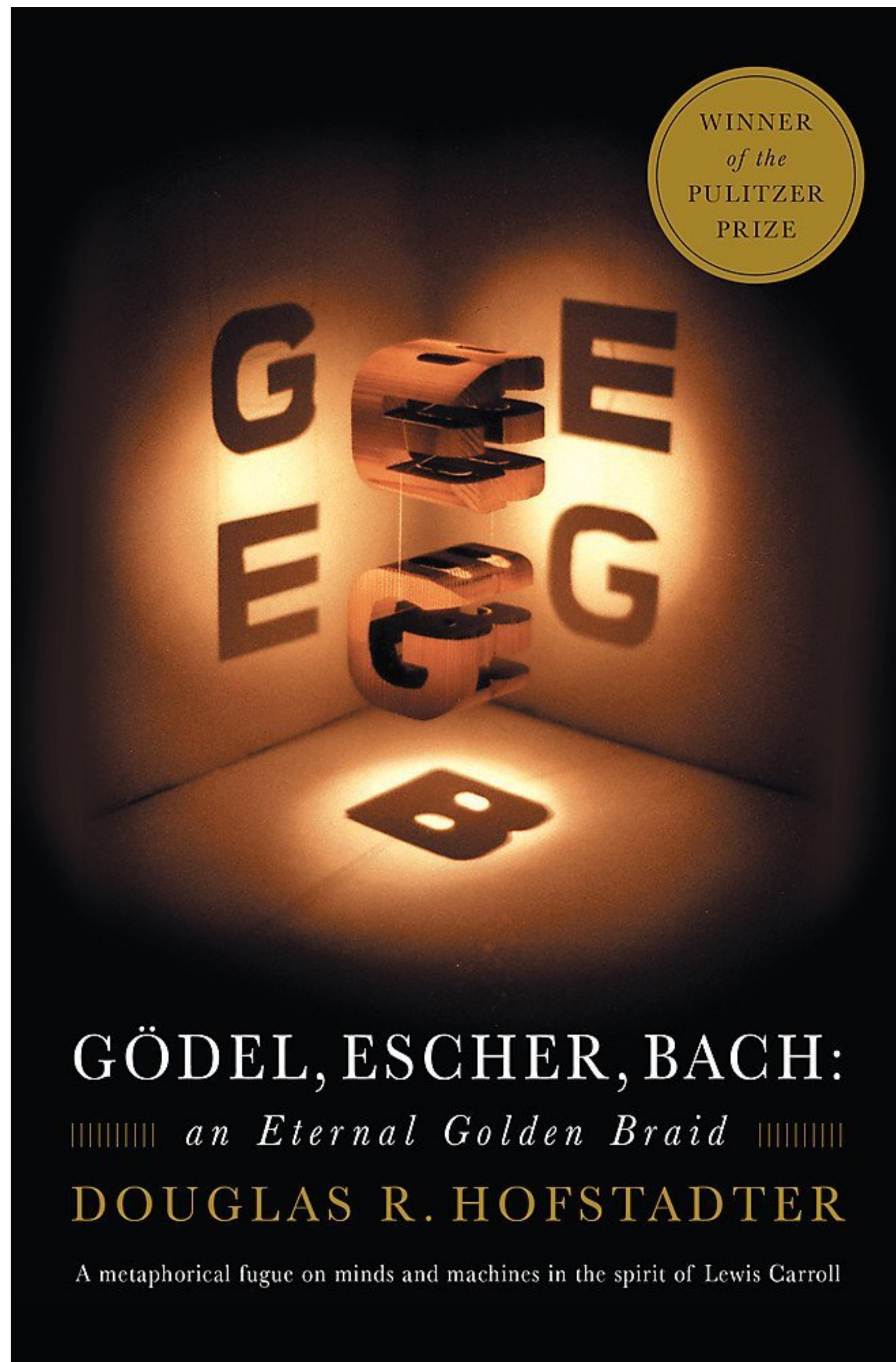
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“Any reachable string must start with **M**”

What is the proof?



MU Puzzle

We have symbols **M**, **I**, and **U**. Start with string **MI**, and transform using the rules:

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2. $Mx \rightarrow Mxx$
3. $xIIIIy \rightarrow xUy$
4. $xUUy \rightarrow xy$

“Any reachable string must start with **M**”

Pre-condition: $A : t^n, t : (\leq)$

selectionSort(int A[], n) {

 i = 0;

 while (i < n - 1) {

 v = i;

 j = i + 1;

 while (j < n) {

 if (A[j] < A[v])

 v = j;

 j++

 }

 swap(A[i], A[v]);

 i++;

 }

 return A;

}

Invariant: $\forall k_1 \forall k_2 . ((0 \leq k_1 < k_2 < n) \wedge (k_1 < i))$
 $\rightarrow (A[k_1] \leq A[k_2])$

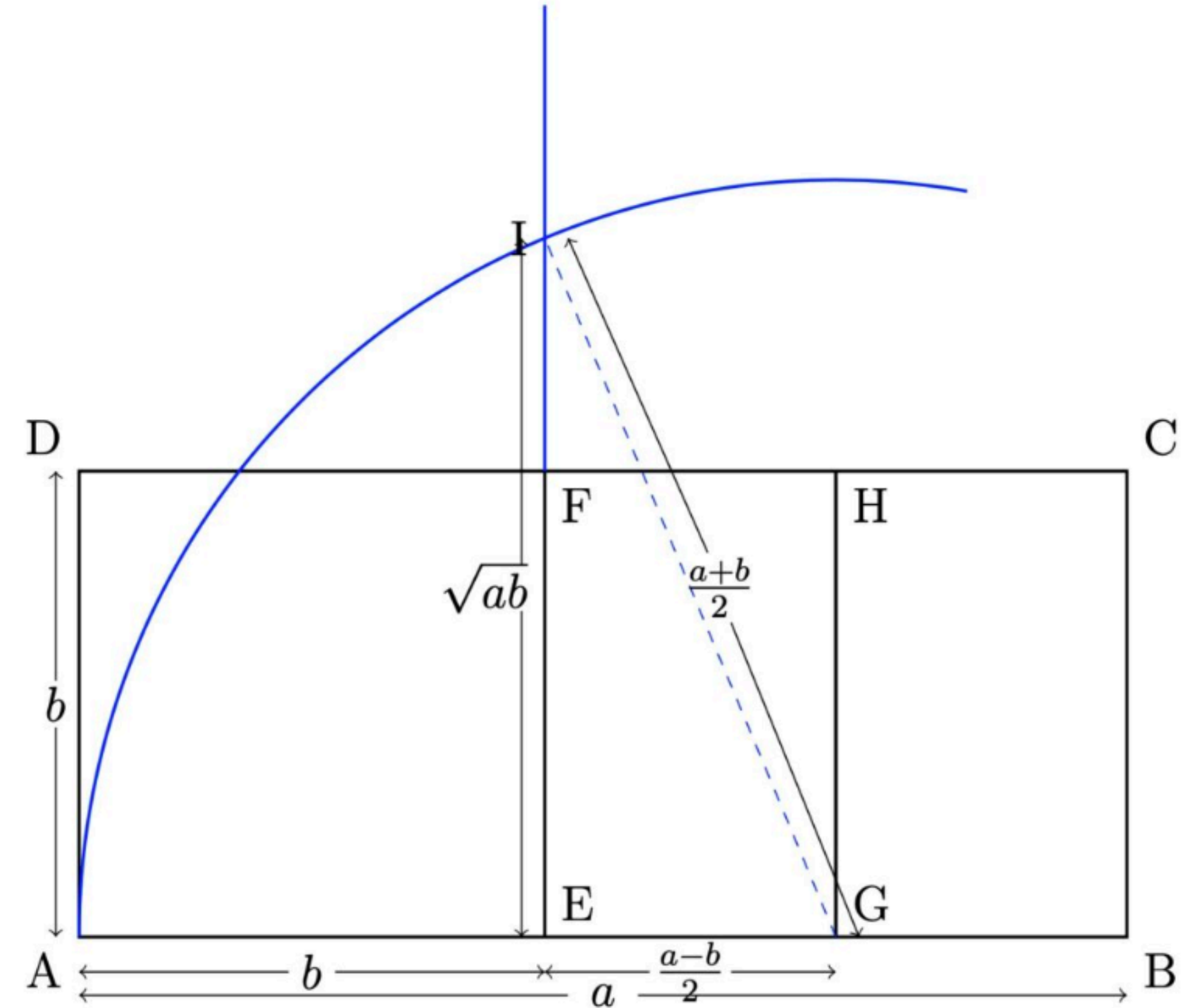
Invariant: $\forall k . (i \leq k \leq j) \rightarrow (A[i] \leq A[k])$

Post-condition: $\forall i \in [n] . \forall j \in [n] . ((i < j) \rightarrow (A[i] \leq A[j]))$



The Invariant is the Proof!

Baudhāyana Śulba (1.54)



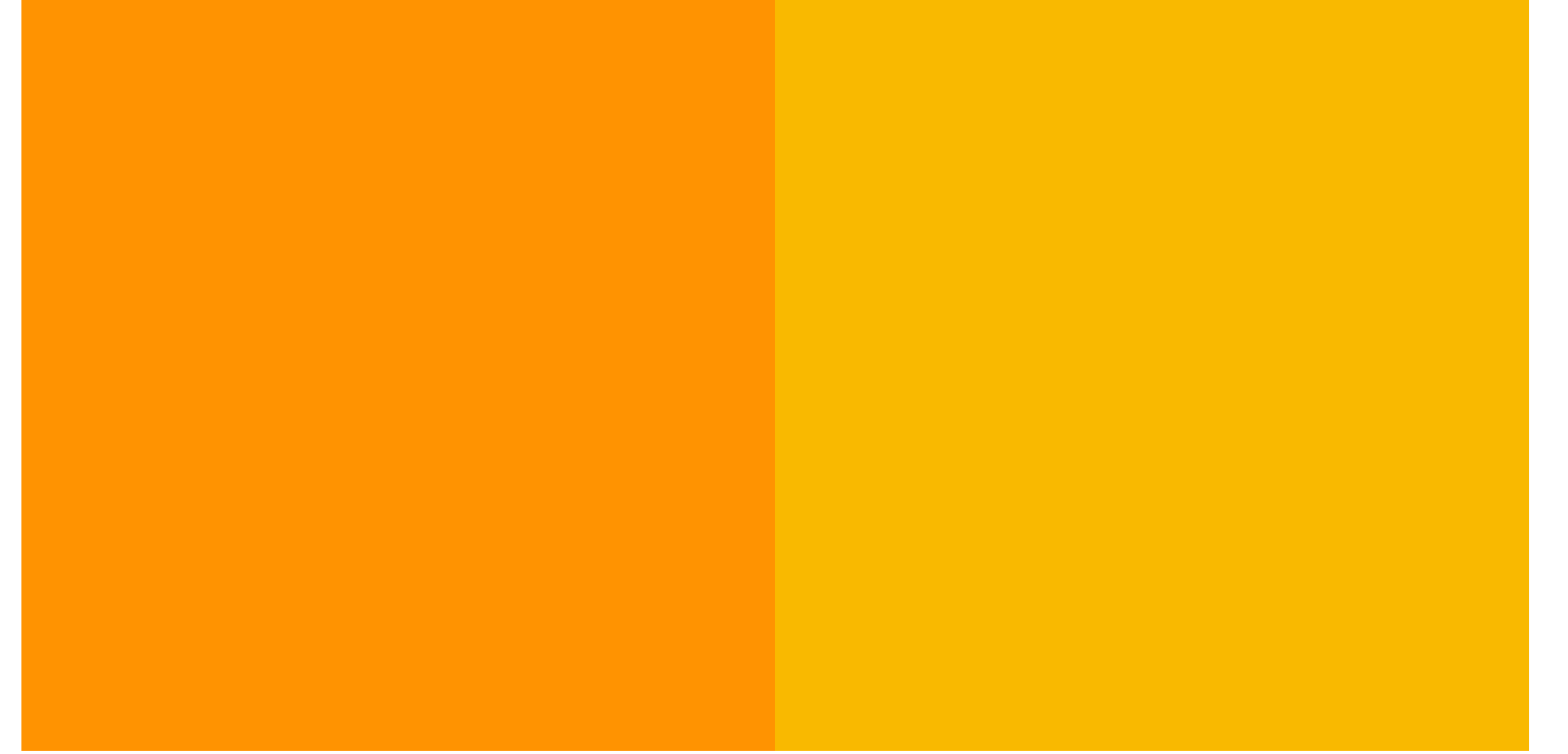
दीर्घचतुरश्रं समचतुरश्रं
चिकीर्षंस्तिर्यङ्मानीं करणीं
कृत्वा शेषं द्वेधा विभज्य
विपर्यस्येतरच्चोपदध्यात् खण्डमावापेन
तत्संपूरयेत् तस्य निर्हारः उत्कः

Baudhāyana Śulba (1.54)



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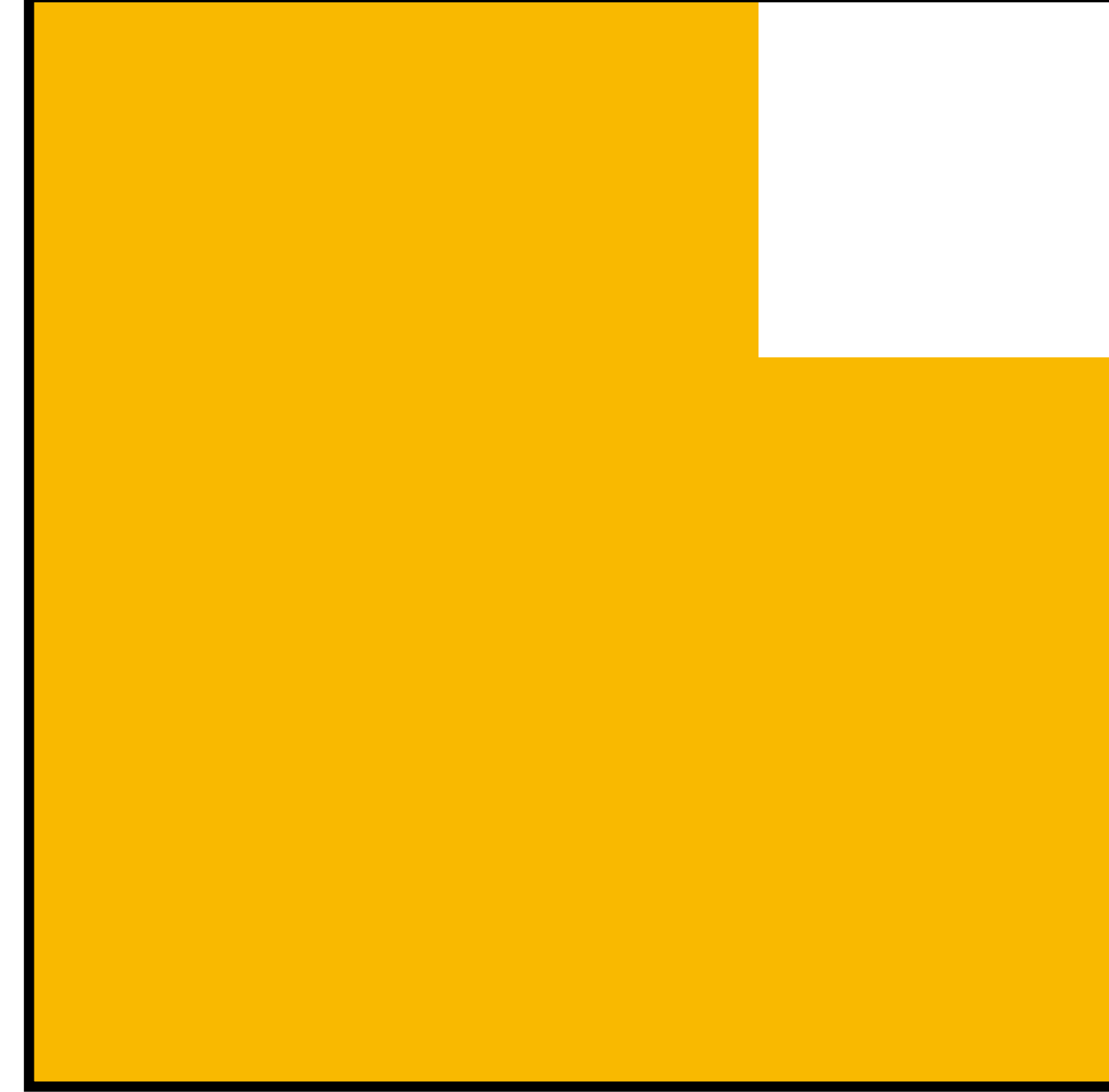
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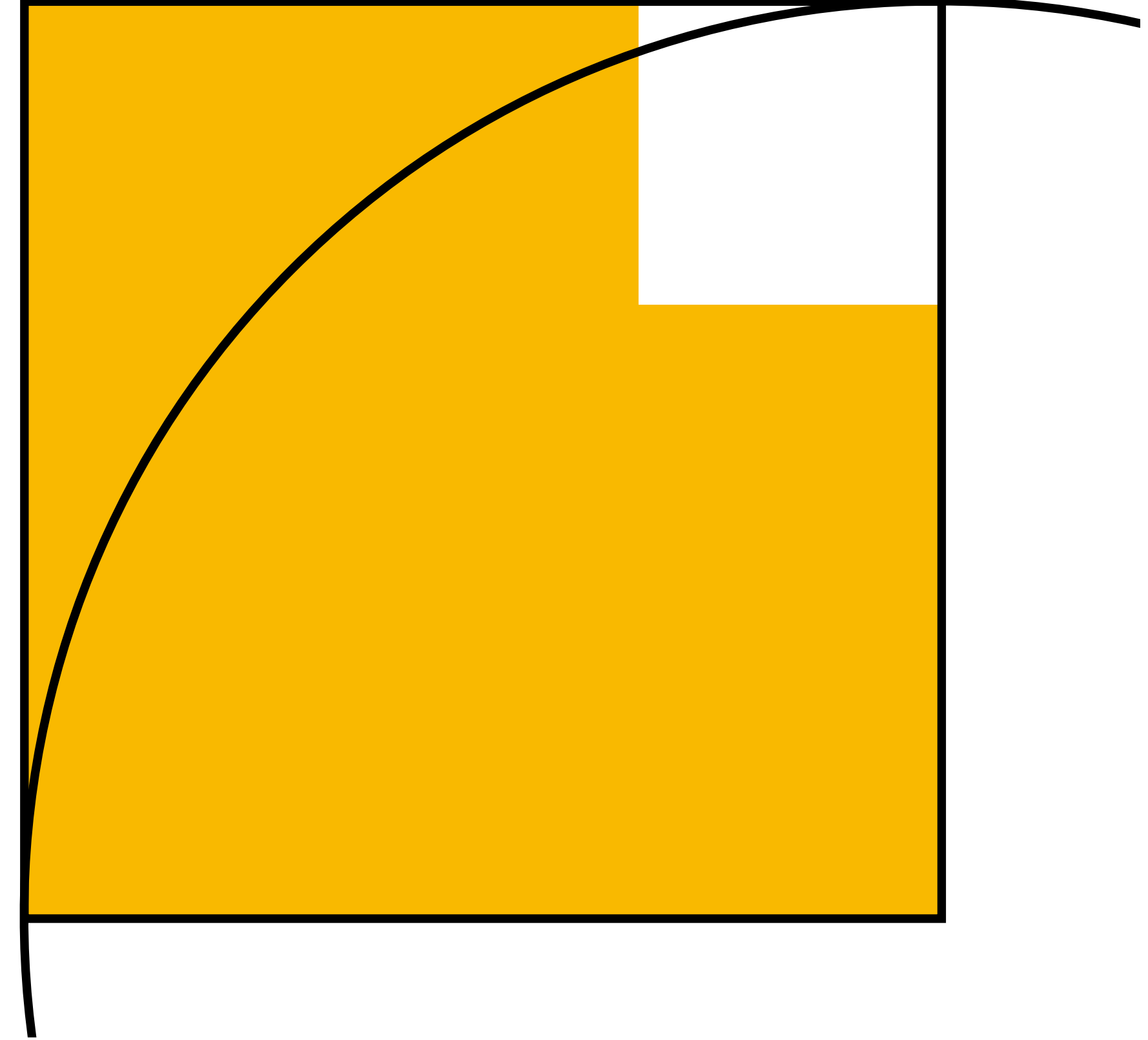
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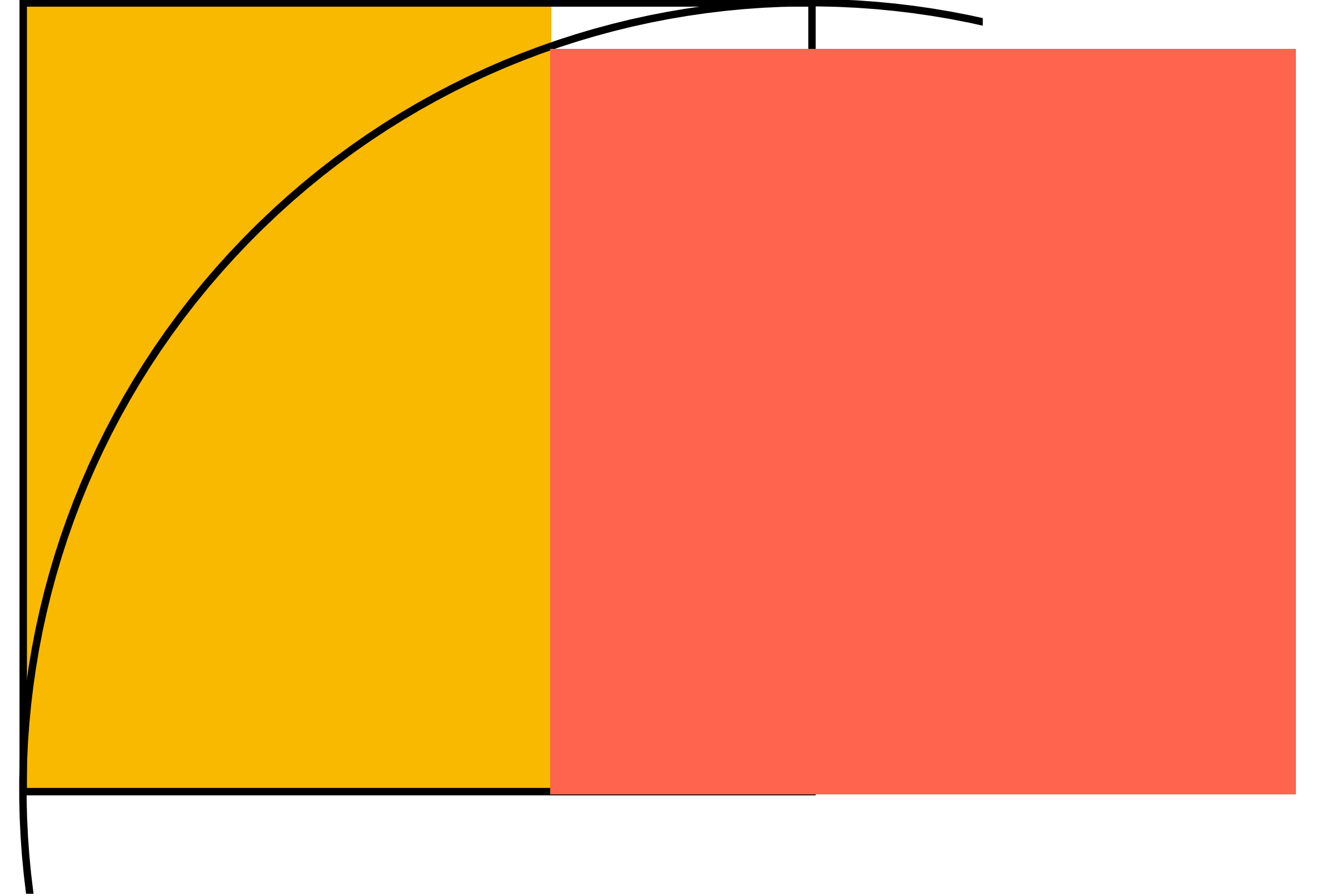
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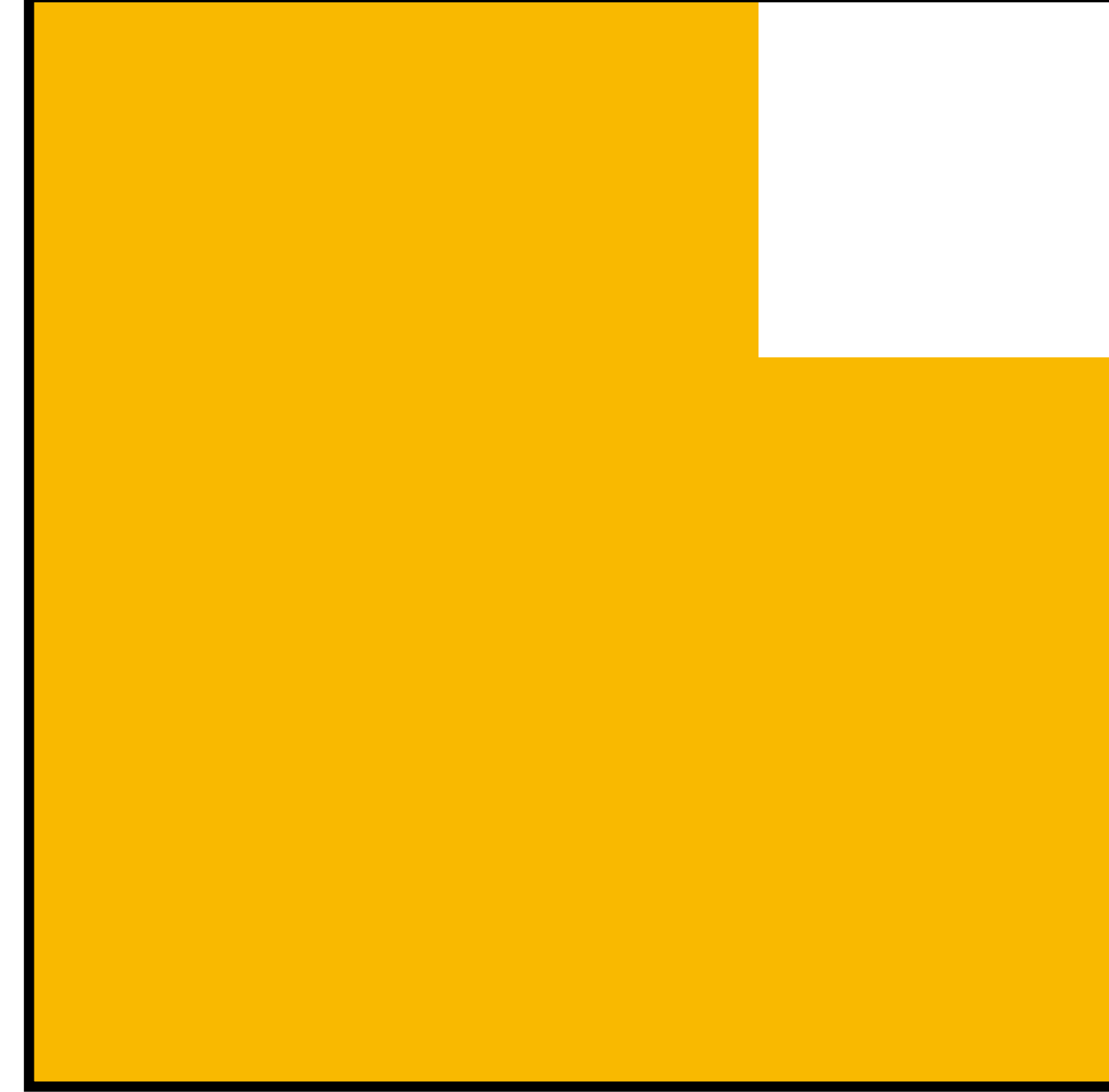
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Baudhāyana Śulba (1.54)



1. A general method to combine squares
2. A general method to subtract squares
3. A general method to convert a rectangle to square
4. Convert n squares to a square

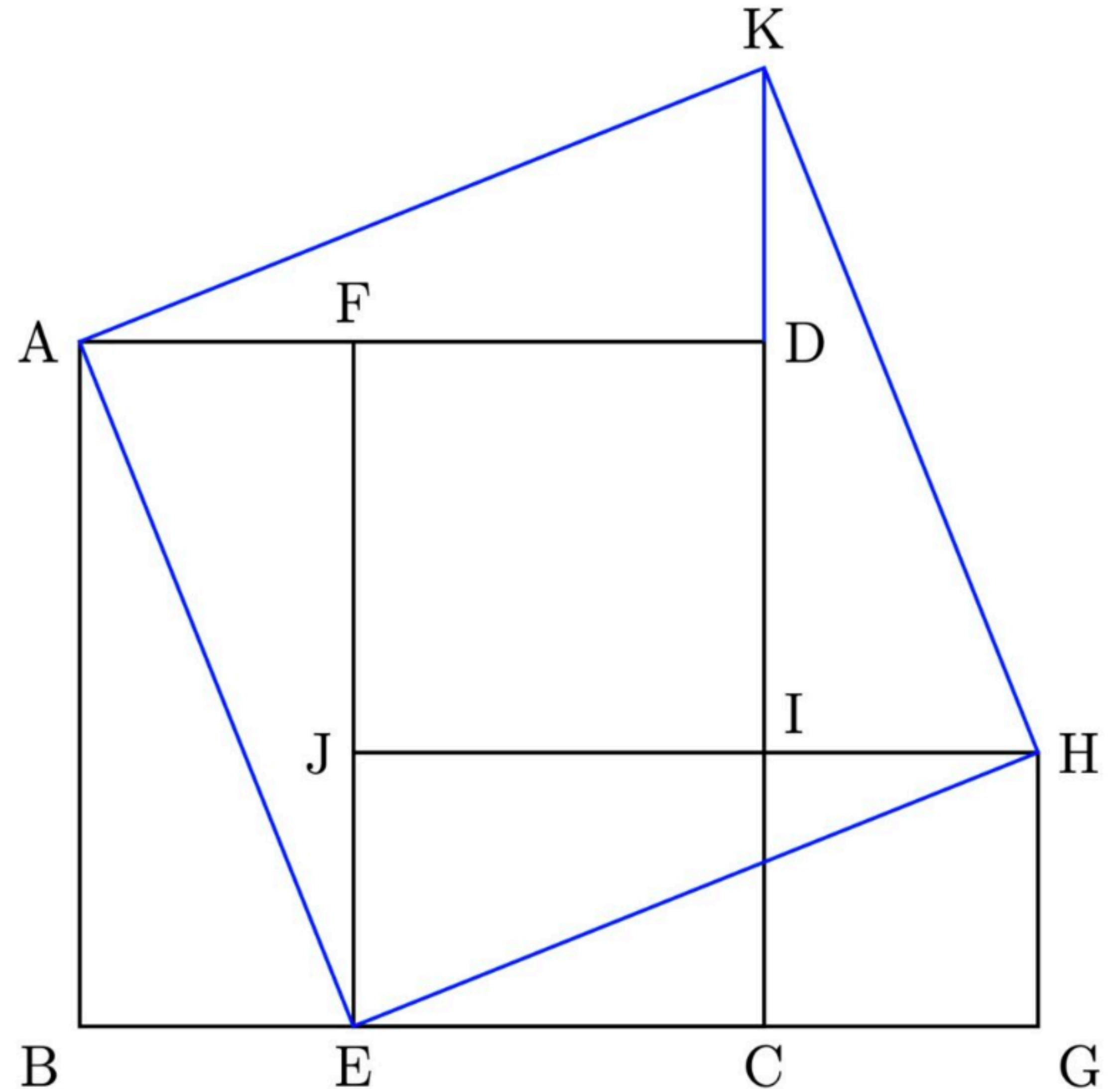
What are the invariants?

A proof of the right-angle theorem:

नानाचतुरश्रेषु समस्यान् कनीयसः
करण्या वर्षीयसो वृद्धम् उल्लिखेत् ।

वृद्धस्य अक्षयारज्जुः समस्तयोः
पार्श्वमानी भवति ॥

Baudhāyana Śulba (1.50)

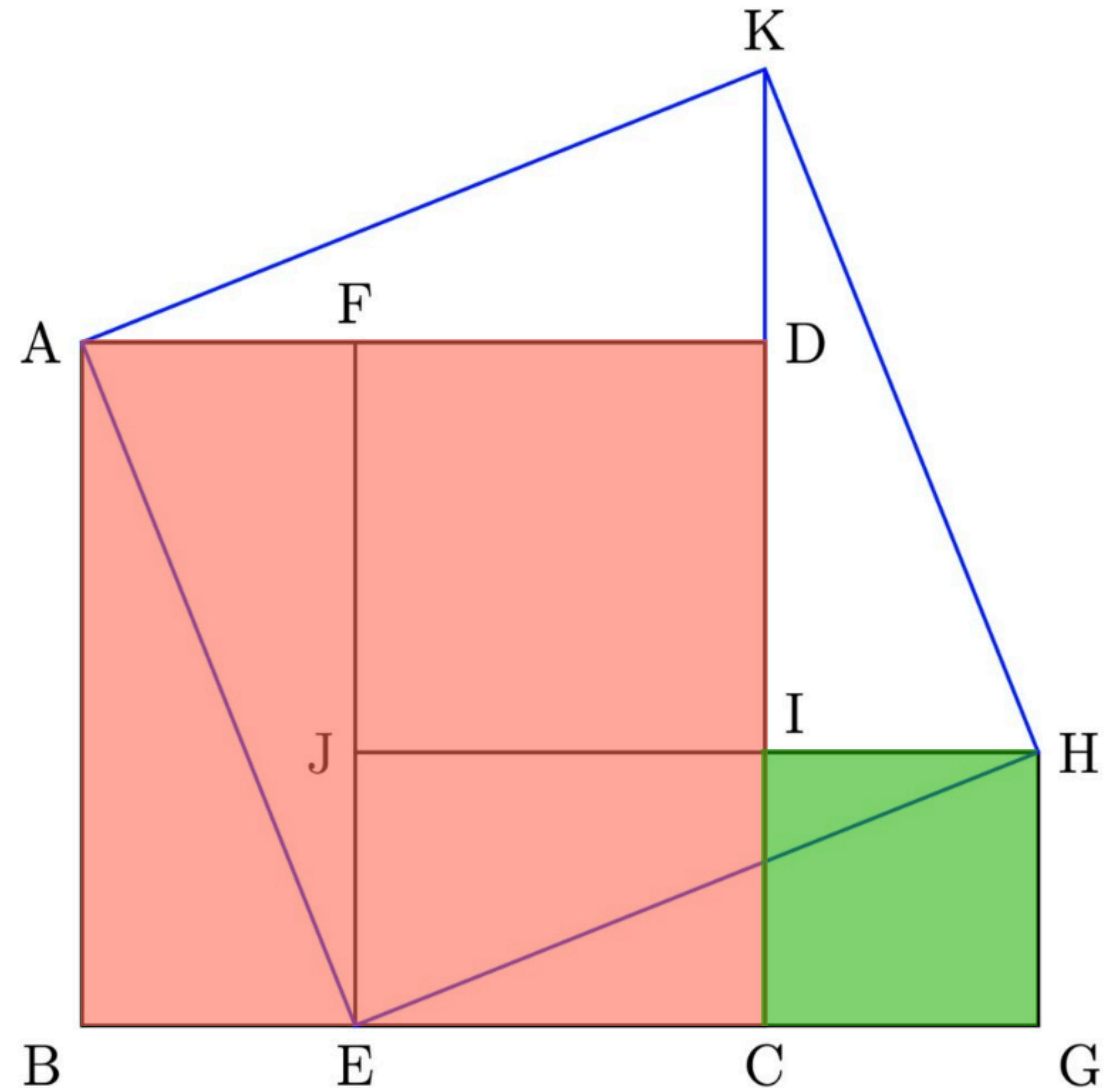


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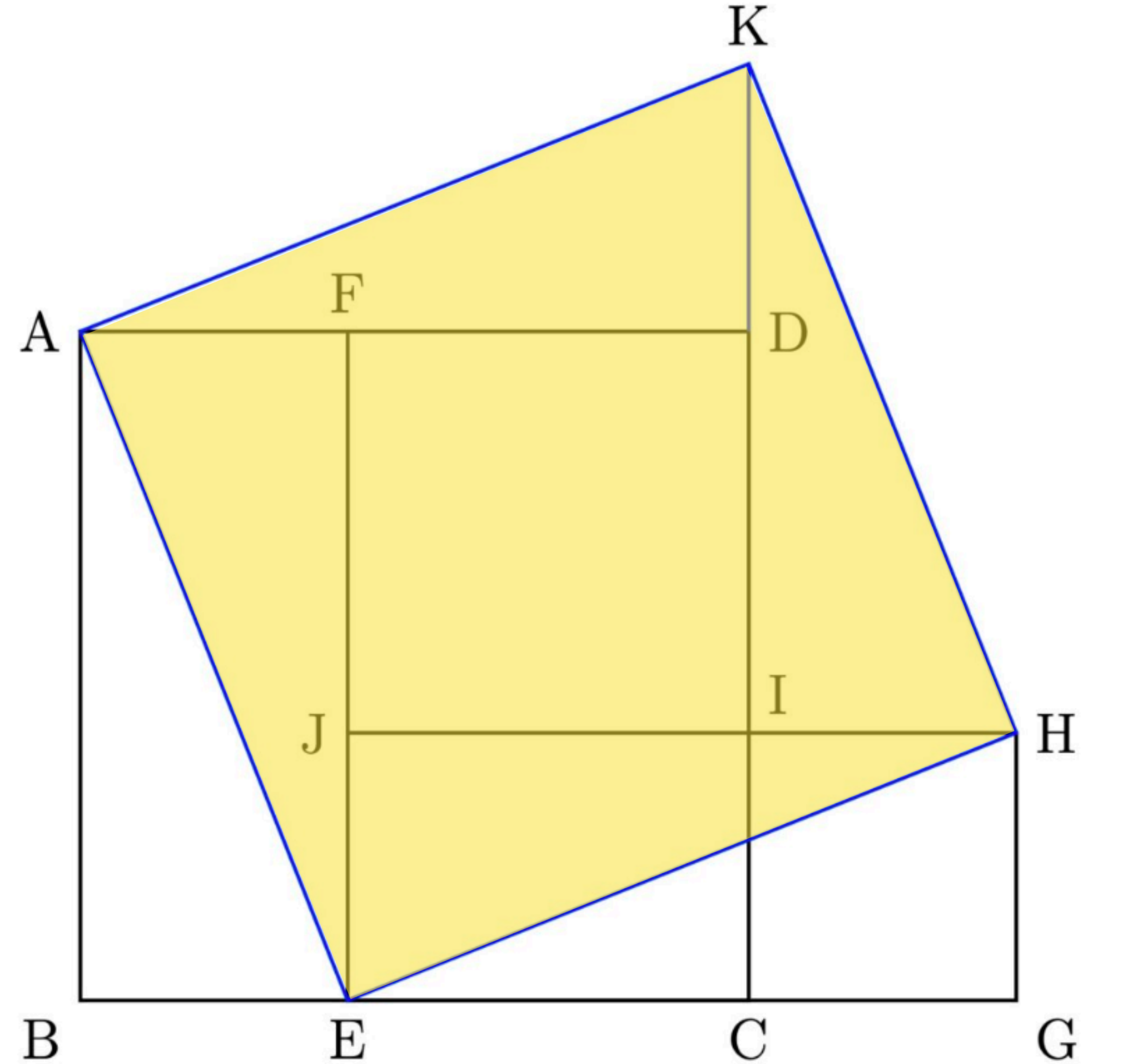


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Baudhāyana Śulba (1.50)



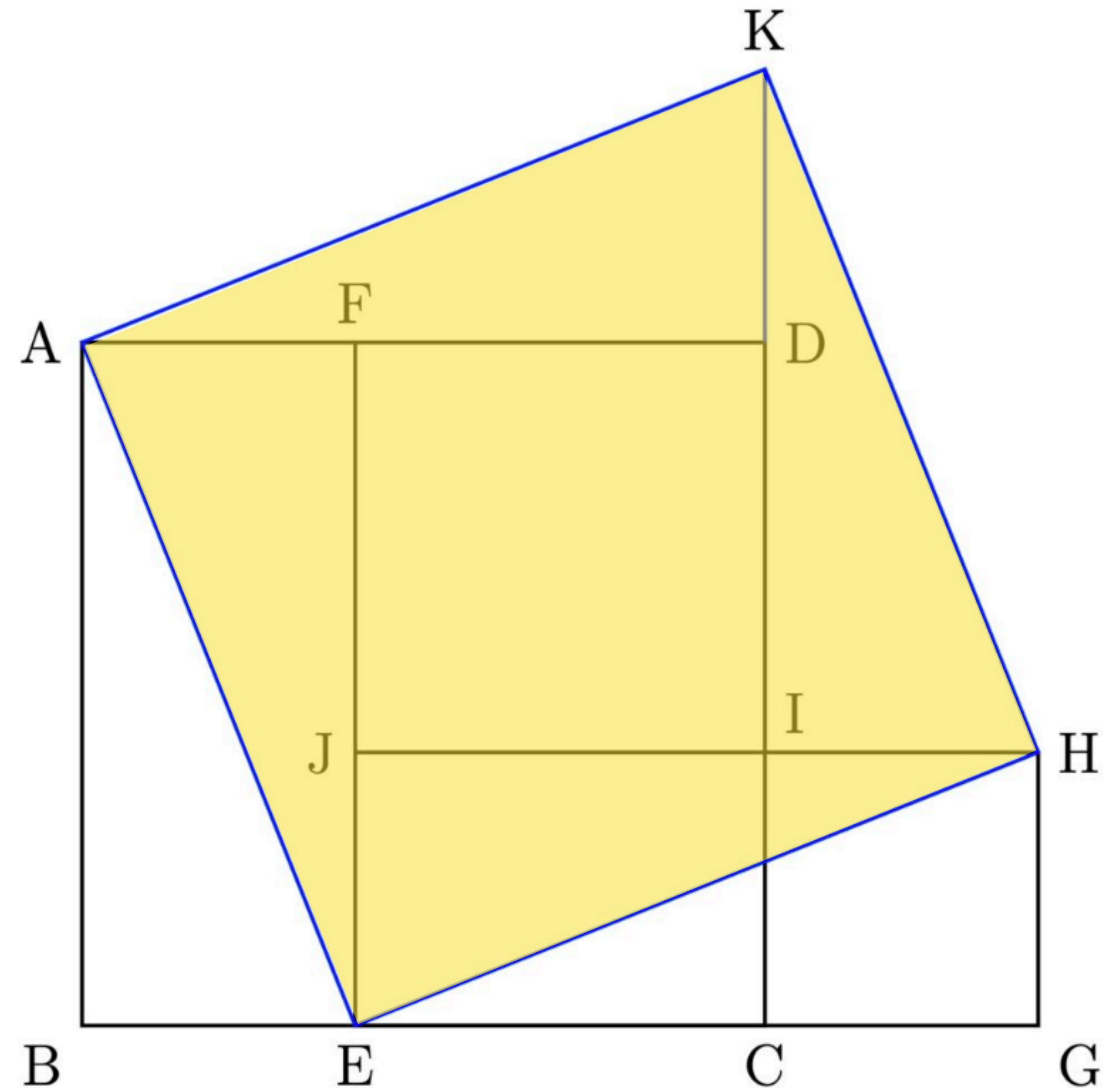
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Baudhāyana Śulba (1.50)

What is the invariant?



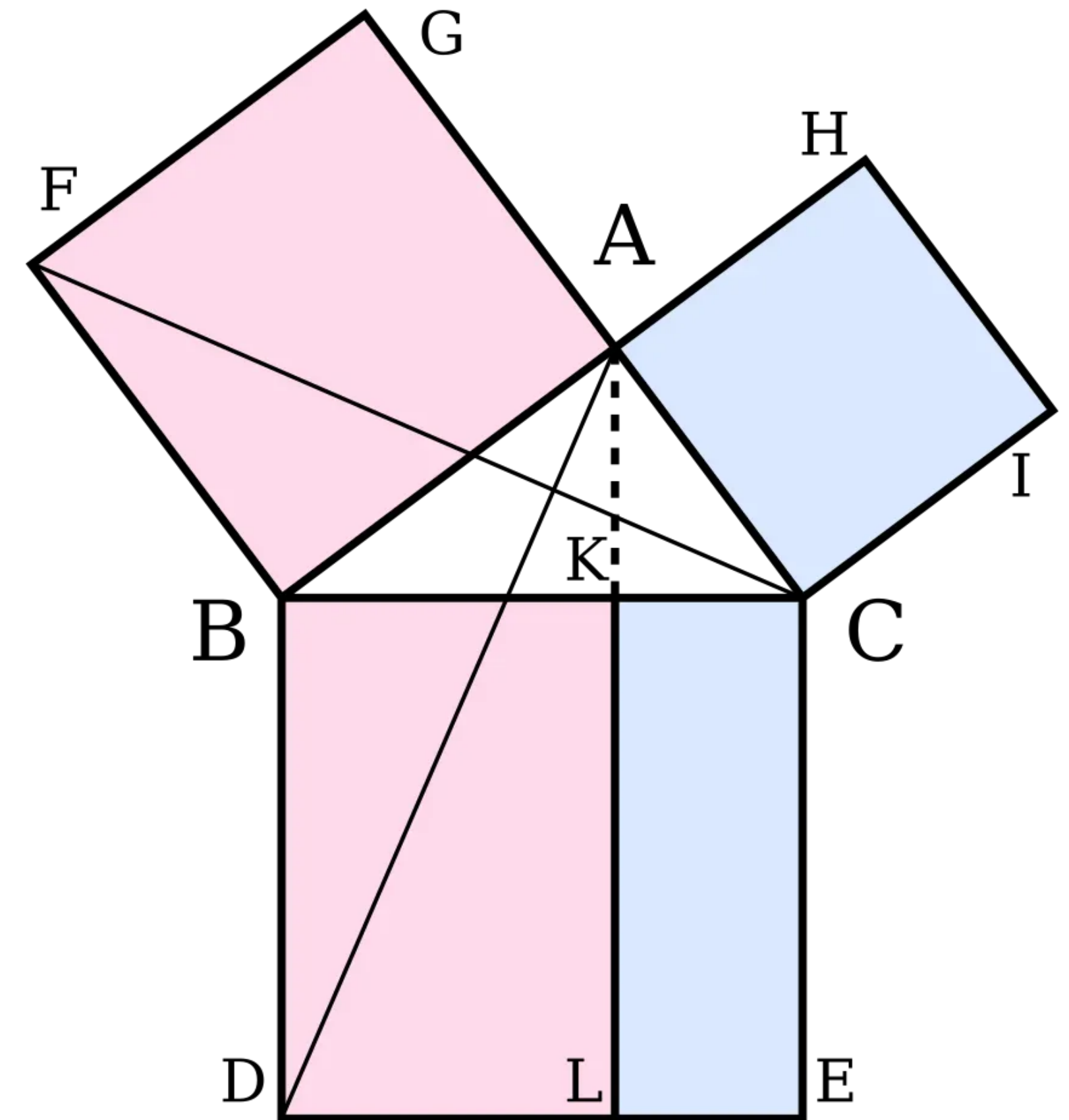


What is a proof?

Postulates:

1. A straight line can be drawn from any point to any point.
2. A finite straight line can be extended indefinitely.
3. A circle with a given radius and a center can be constructed.
4. All right angles are equal.
5. Through a point, there is exactly one line parallel to the given line.

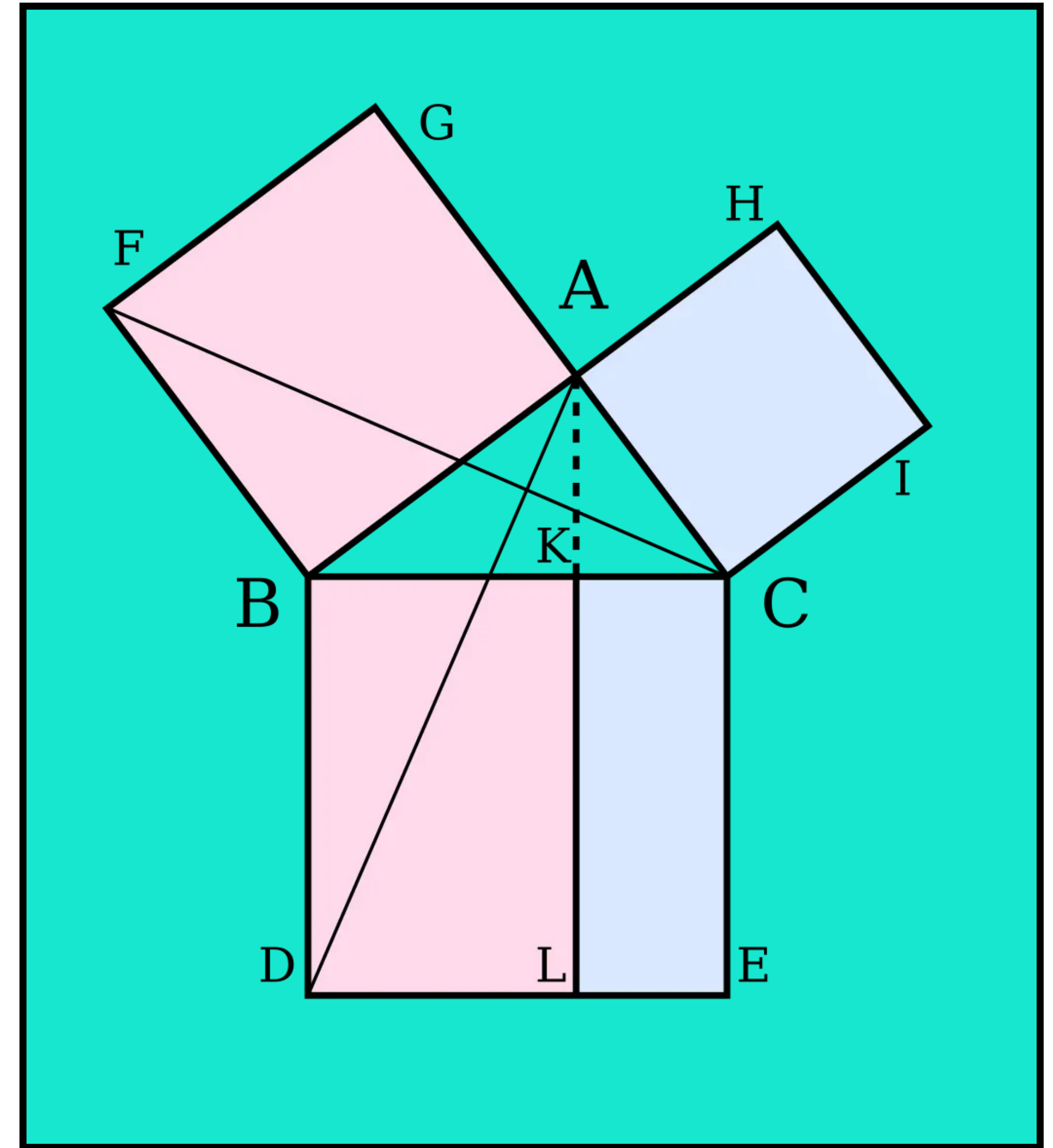
Euclid's Elements (I, 47)

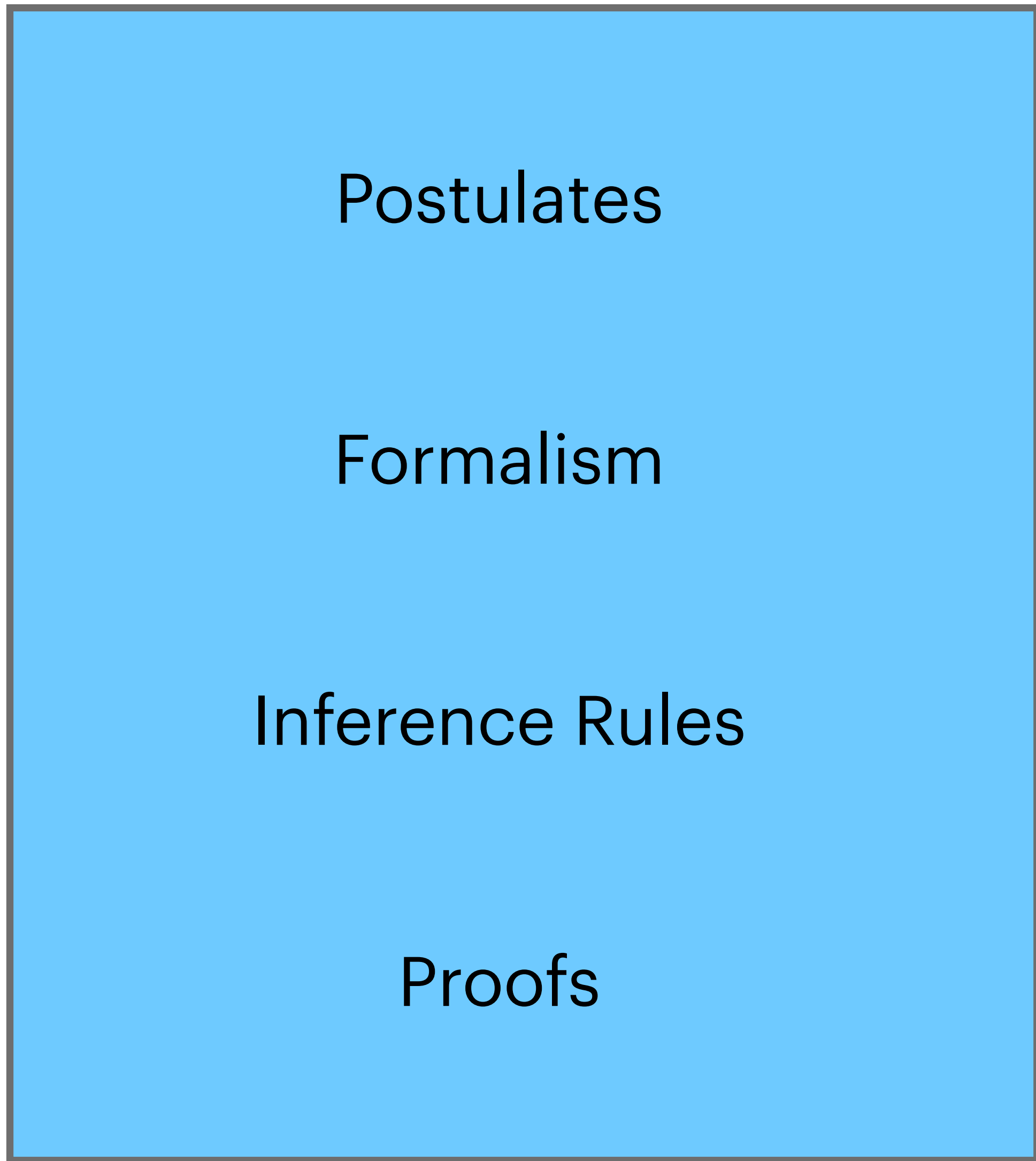


Postulates:

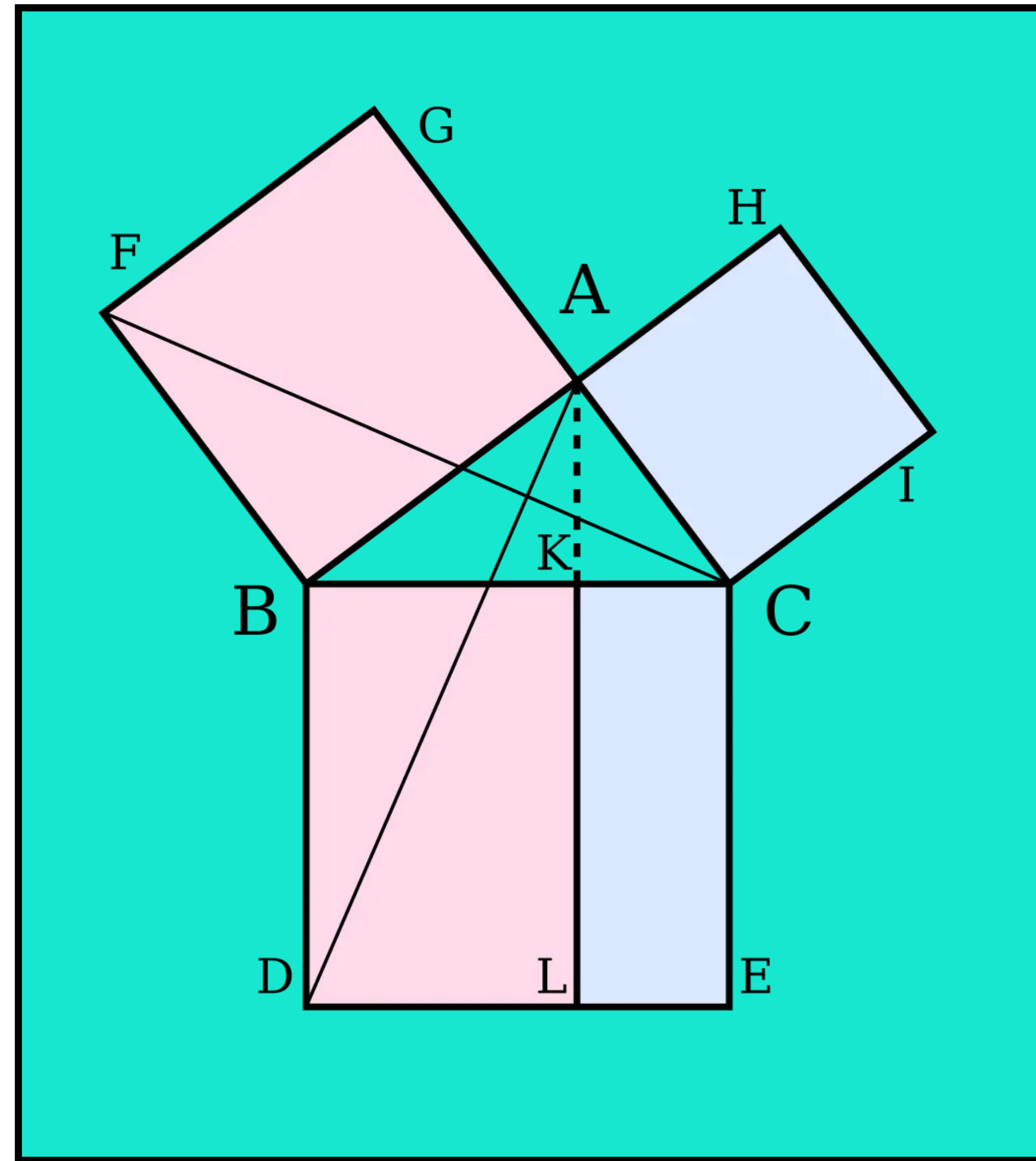
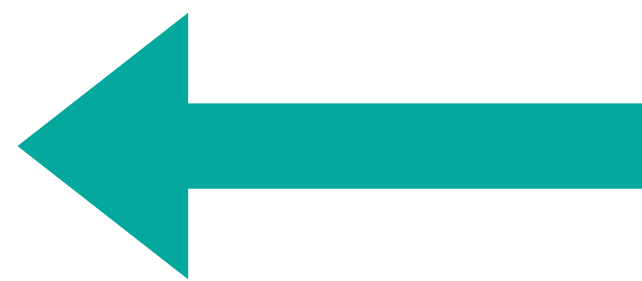
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Euclid's Elements (I, 47)





Abstraction



Postulates

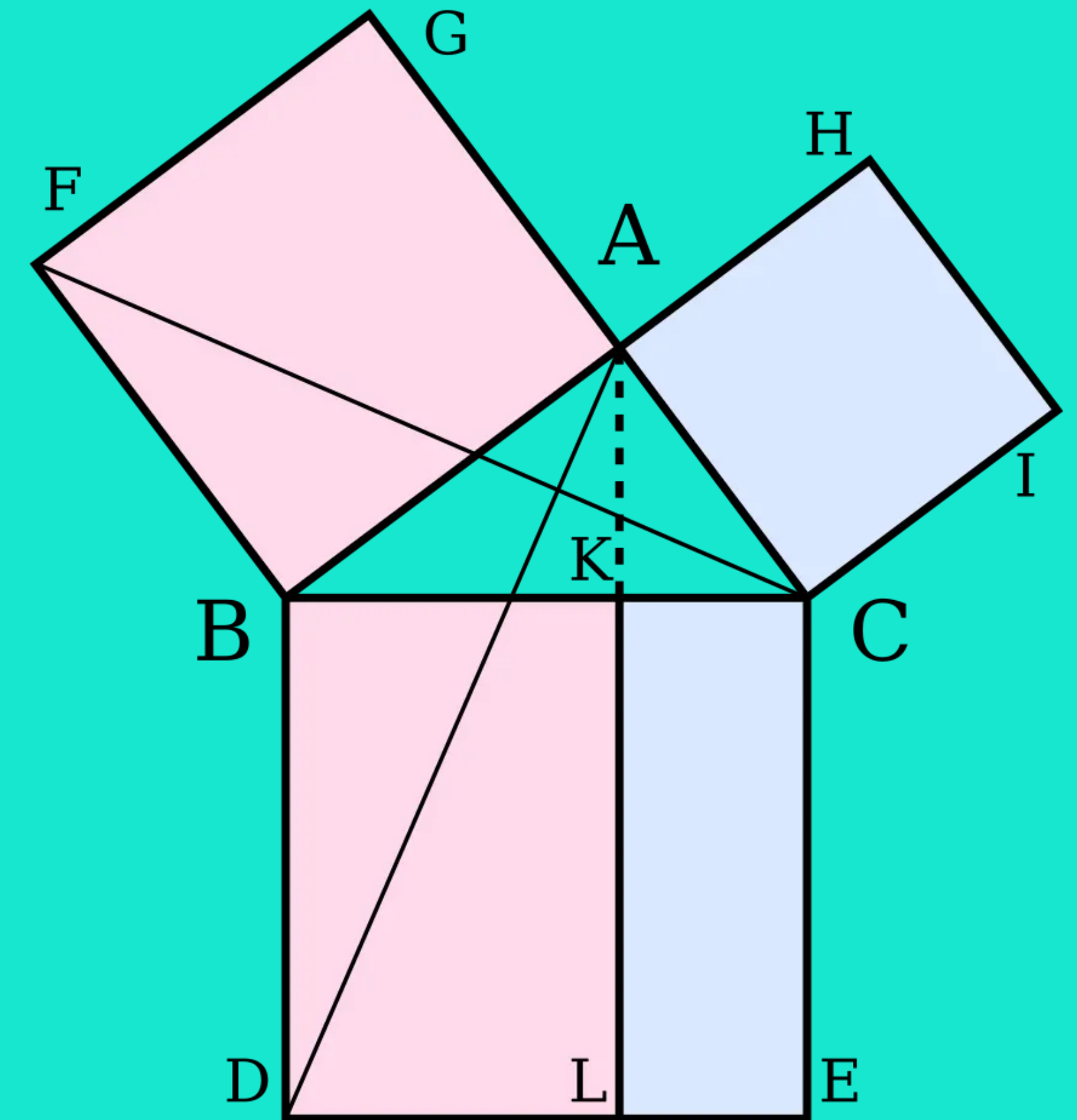
Formalism

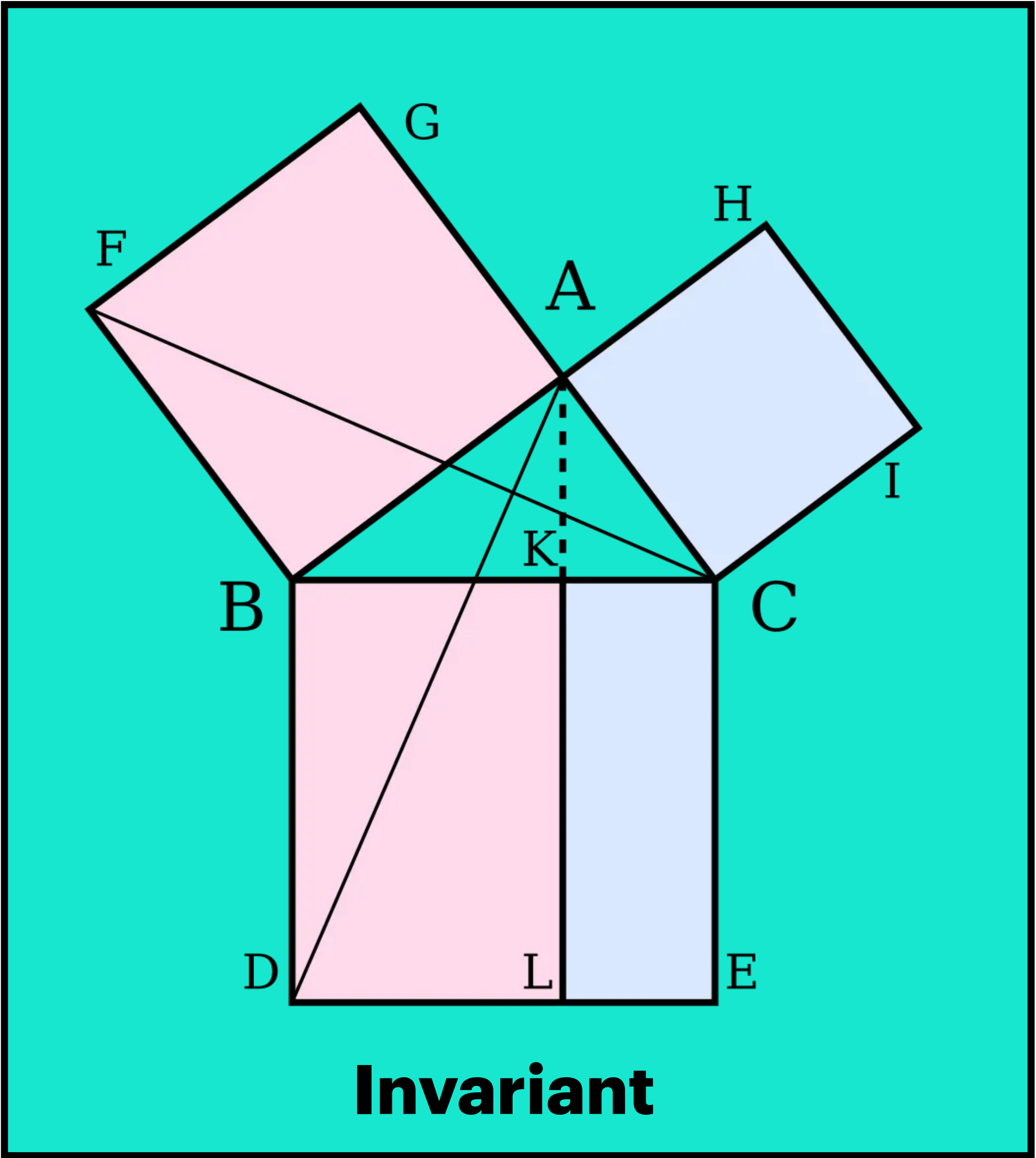
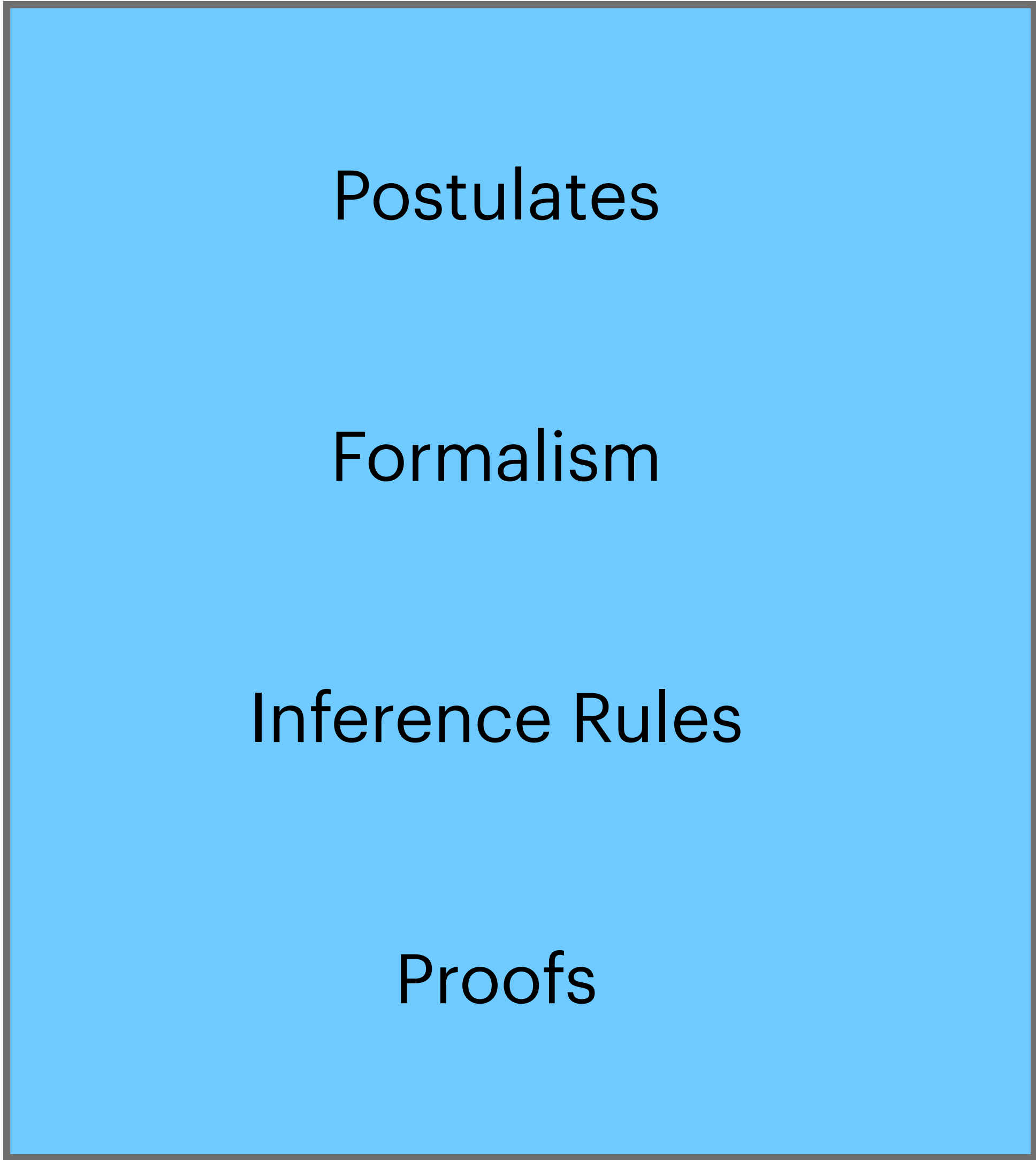
Inference Rules

Proofs

Abstraction

??





Baudhāyana Śulba (1.50)

Does a formal proof have invariants too?

Postulates

Formalism

Inference Rules

Proofs

Does a formal proof have invariants too?

Postulates

Formalism

Inference Rules

Proofs

$$P \wedge Q \vdash Q \wedge P$$

$$\frac{\frac{P \wedge Q}{Q} (\wedge E)}{\frac{Q \wedge P}{P} (\wedge I)} (\wedge E)$$

Does a formal proof have invariants too?

Postulates

Formalism

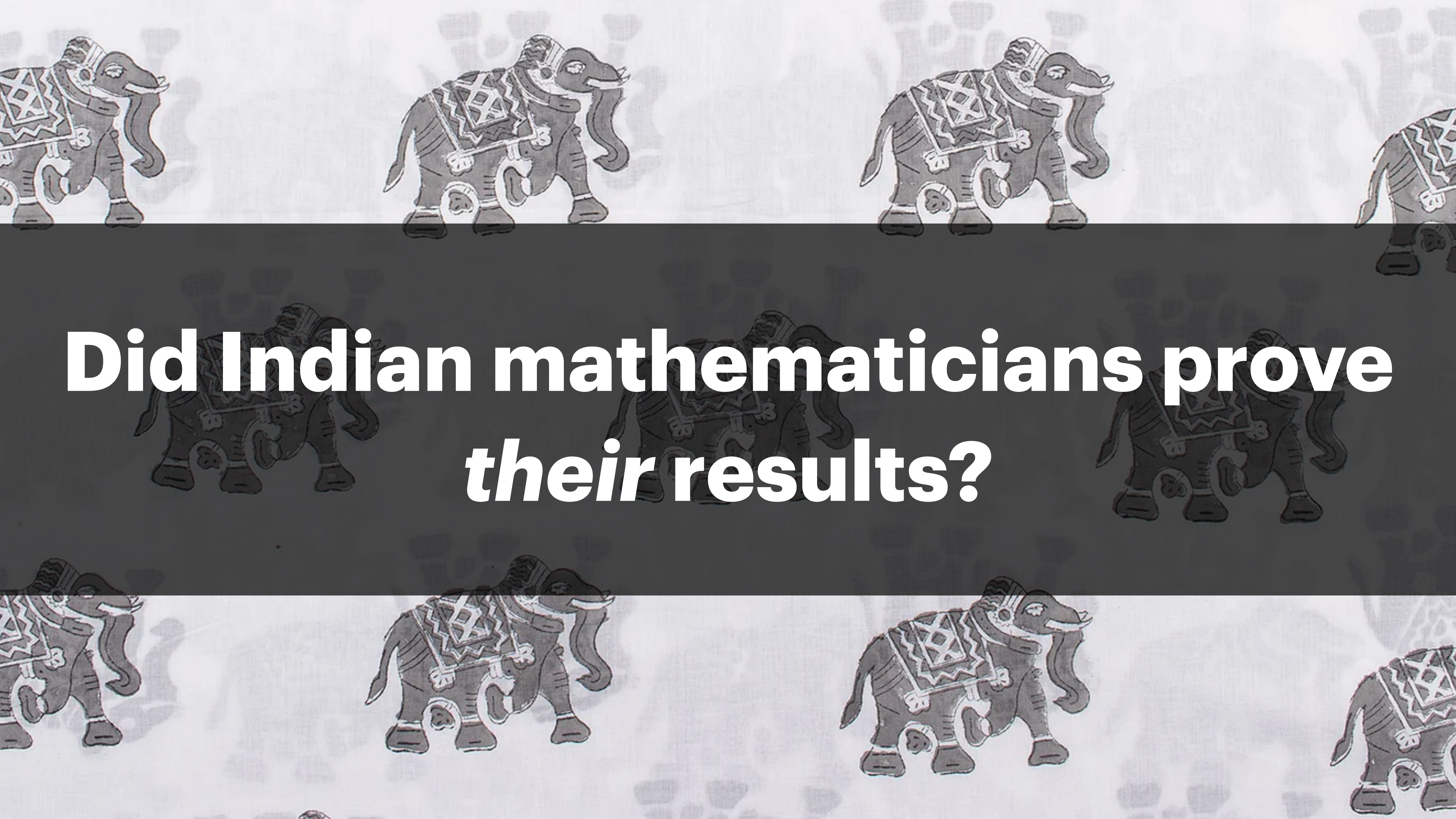
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Proofs

$$P \wedge Q \vdash Q \wedge P$$

$$\frac{\frac{P \wedge Q}{Q} (\wedge E)}{\frac{Q \wedge P}{P} (\wedge I)} (\wedge E)$$

Validity is the invariant



**Did Indian mathematicians prove
their results?**

Deficiency View

Formal Greco-European Proofs of Correctness

No

Corrective View

Upapatti in commentaries

Yes, explicitly.

Deficiency View

Formal Greco-
European Proofs of
Correctness

No

Corrective View

Upapatti in
commentaries

Yes, explicitly.

Invariant View

In the primary text?

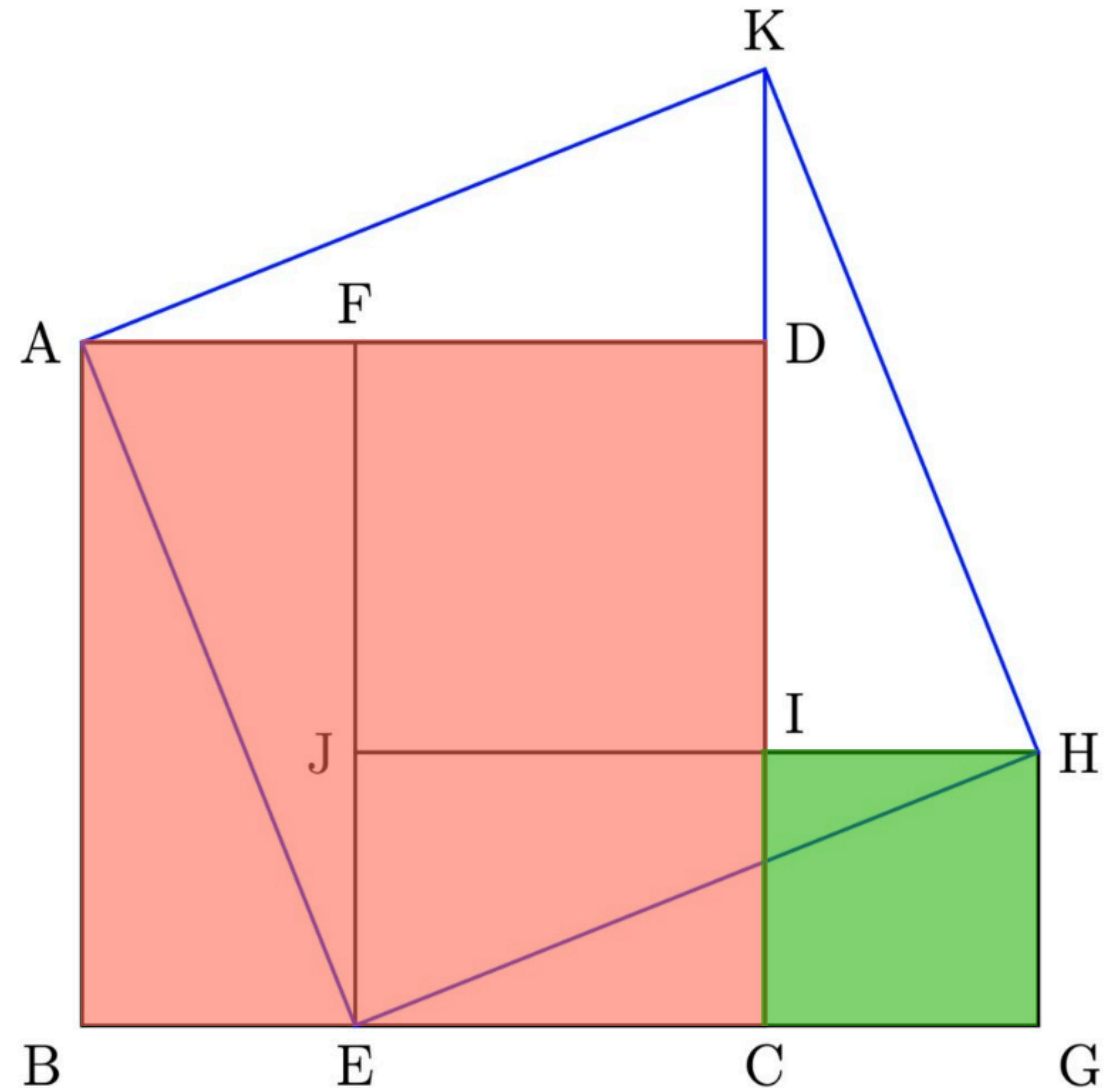
Yes, implicitly.

A proof of the right-angle theorem:

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पार्श्वमानी भवति ॥

Baudhāyana Śulba (1.50)



For the Indians, of course, the effectiveness of the *cakravāla* could be no more than an experimental fact, based on their treatment of great many specific cases, some of them of considerable complexity and involving (to their delight, no doubt) quite large numbers. As we shall see, Fermat was the first one to perceive the need for a general proof, and Lagrange was the first to publish one. Nevertheless, to have developed the *cakravāla* and to have applied it successfully to such difficult numerical cases as $N = 61$, or $N = 67$ had been no mean achievements.

Bhāvanā (Composition)

If (x_1, y_1, z_1) and (x_2, y_2, z_2) are solutions to $Dx^2 + z = y^2$, then so is:

$$(x_1y_2 + x_2y_1, y_1y_2 + Dx_1x_2, z_1z_2)$$

$$(x_1, y_1, z_1) \circ (x_2, y_2, z_2) = (x_1y_2 + x_2y_1, y_1y_2 + Dx_1x_2, z_1z_2)$$



Brahmagupta (c. 598 – c. 668 CE)

Bhāvanā (Composition)

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$$\text{Invariant: } Dx^2 + z = y^2$$



Brahmagupta (c. 598 – c. 668 CE)

Ramanujan's Master Theorem

If a complex-valued function $f(x)$ has an expansion of the form

$$f(x) = \sum_{k=0}^{\infty} \frac{\varphi(k)}{k!} (-x)^k,$$

where $\varphi(s)$ is an analytic function, then the Mellin transform of $f(x)$ is given by

$$\int_0^{\infty} x^{s-1} f(x) dx = \Gamma(s) \varphi(-s),$$

where $\Gamma(s)$ is the gamma function.

11. $\int_0^{\infty} e^{-x} x^n dx = \Gamma(n+1)$ and hence

$$\int_0^{\infty} x^{n-1} \left\{ \phi(0) - \frac{x}{1!} \phi(1) + \frac{x^2}{2!} \phi(2) - \dots \right\} dx = \Gamma(n) \phi(-n)$$

Sol. $\int_0^{\infty} e^{-x} x^n dx = e^{-x} \{ x^n + n x^{n-1} + n(n-1) x^{n-2} + \dots \}$
 when $x=0 = \Gamma(n+1)$ by IV 10 Cor.

$$f(0) \int_0^{\infty} e^{-x} x^{n-1} dx = \Gamma(n) f(0)$$

$$\frac{f'(0)}{1!} \int_0^{\infty} e^{-x} x^n dx = \frac{\Gamma(n)}{1!} \frac{f'(0)}{1!}$$

$$\frac{f''(0)}{2!} \int_0^{\infty} e^{-x} x^{n+1} dx = \frac{\Gamma(n+1)}{2!} \frac{f''(0)}{2!}$$

and so on.

Adding up all the results we have

$$\int_0^{\infty} x^{n-1} \left\{ f(0) - \frac{x}{1!} f(1) + \frac{x^2}{2!} f(2) - \dots \right\} dx = \Gamma(n) f\left(-\frac{n}{2}\right)$$

Let $f(x^n) = \phi(n)$ then $f\left(\frac{1}{x^n}\right) = \phi(-n)$.

Cor 1. $\int_0^{\infty} x^{n-1} \left\{ \phi(0) - x \phi(1) + x^2 \phi(2) - \dots \right\} dx = \frac{\pi \phi(-n)}{\sin \pi n}$

Cor 2. $\int_0^{\infty} x^{n-1} \left\{ \phi(0) - \frac{x^2}{2!} \phi(2) + \frac{x^4}{4!} \phi(4) - \dots \right\} dx = \frac{\Gamma(n) \phi(-n)}{x \cos \frac{\pi n}{2}}$

Cor 3. $\int_0^{\infty} \left\{ \phi(0) - \frac{x}{1!} \phi(1) + \frac{x^2}{2!} \phi(2) - \dots \right\} \cos \pi x dx$
 $= \phi(-1) - \pi^2 \phi(-3) + \pi^4 \phi(-5) - \dots$

Cor 4. $\int_0^{\infty} \left\{ \phi(0) - x^2 \phi(2) + x^4 \phi(4) - \dots \right\} \cos \pi x dx$
 $= \frac{\pi}{2} \left\{ \phi(-1) - \frac{\pi}{1!} \phi(-2) + \frac{\pi^2}{2!} \phi(-3) - \frac{\pi^3}{3!} \phi(-4) + \dots \right\}$

Curry–Howard Correspondence

A proposition can be viewed as a type, and a proof of that proposition as a program (term) inhabiting that type.

A proof of $(A \Rightarrow B)$ is a function that transforms proofs of (A) into proofs of (B) :

$$A \Rightarrow B \longleftrightarrow A \rightarrow B$$

$$\frac{\Gamma, x \vdash b}{\Gamma \vdash \lambda x. b \rightarrow B}$$

function `f (x : A) : B`

projectbrahmagupta.github.io

Problems from *Indian mathematics.*

Project Brahmagupta is an archive of १०८ problems in mathematics and computation, drawn from and inspired by the works of Indian mathematicians.

Contribute a problem

१	२	३	४	५	६	७	८	९	१०	११	१२
१३	१४	१५	१६	१७	१८	१९	२०	२१	२२	२३	२४
२५	२६	२७	२८	२९	३०	३१	३२	३३	३४	३५	३६
३७	३८	३९	४०	४१	४२	४३	४४	४५	४६	४७	४८
४९	५०	५१	५२	५३	५४	५५	५६	५७	५८	५९	६०
६१	६२	६३	६४	६५	६६	६७	६८	६९	७०	७१	७२
७३	७४	७५	७६	७७	७८	७९	८०	८१	८२	८३	८४
८५	८६	८७	८८	८९	९०	९१	९२	९३	९४	९५	९६
९७	९८	९९	१००	१०१	१०२	१०३	१०४	१०५	१०६	१०७	१०८